Name	Subject Class	Class	Candidate Number
	2BIX01		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY **8875/02**
25 AUGUST 2014
HIGHER 1 **2 hours**

Paper 2

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST
Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5 or 6	
Total	**60**

This question paper consists of **10** printed pages.

[Turn over

1 Fig. 1.1 is an electron micrograph of part of a chloroplast in a plant cell.

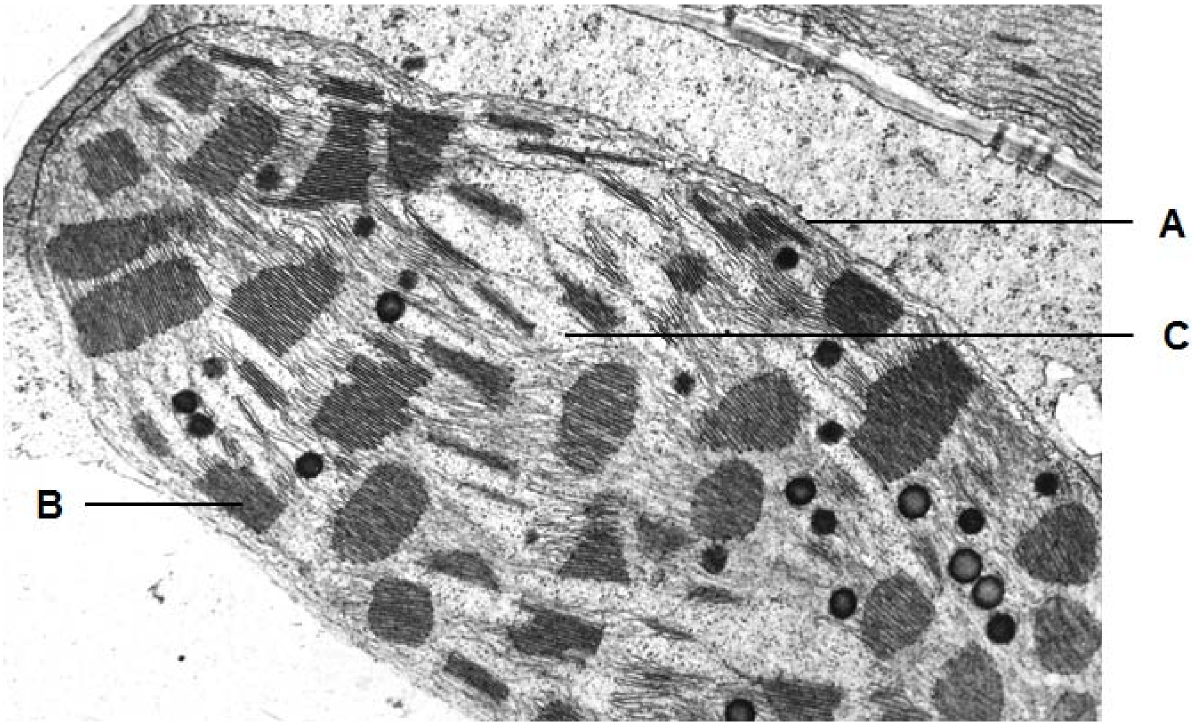

Fig. 1.1

(a) Explain **two** features of starch that allow it to be stored within the chloroplast.

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....

....

.... [2]

(b) State **one** similarity and **one** difference between structures **A** and **B**.

....

....

....

.... [2]

For examiner's use

(c) Describe how ATP is synthesised in the light dependent reaction.

[3]

(d) The enzyme Rubisco is found in **C**. Suggest and explain the effect on the rate of light-dependent reactions if the gene coding for the enzyme Rubisco is mutated.

[3]

[Total: 10]

2 Fig. 2.1 shows the structure of molecules and some associated processes found in a cell.

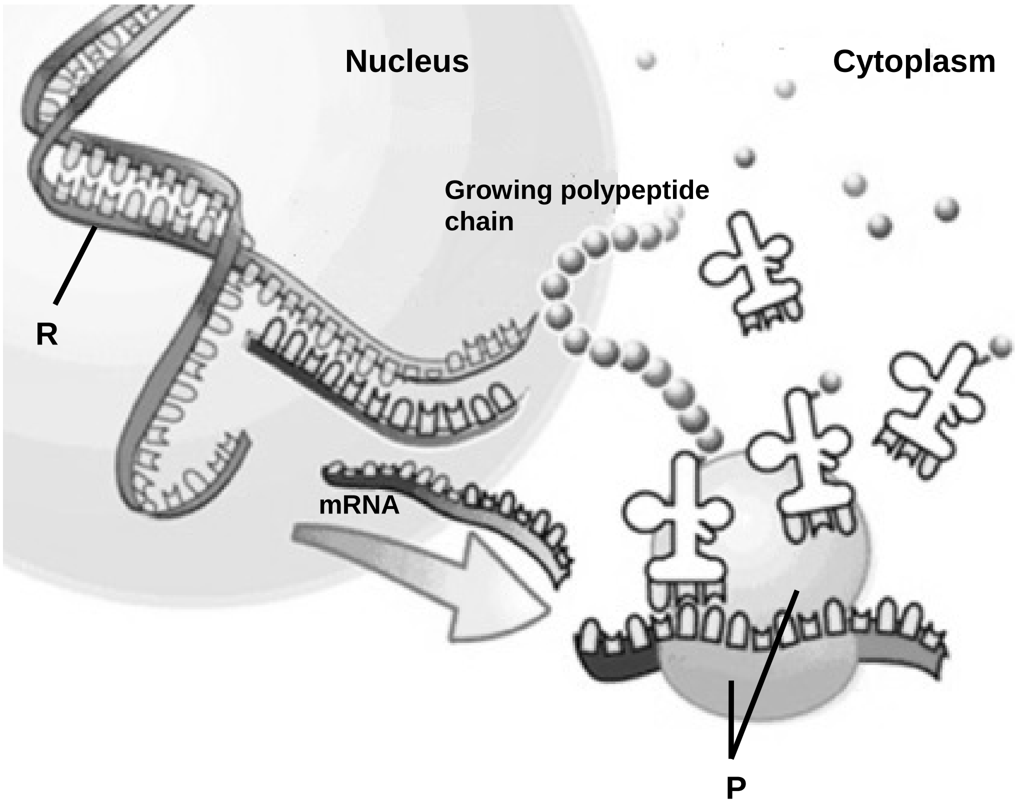

Fig. 2.1

(a) (i) Name the type of bond holding the two strands of **R** together.

.... [1]

(ii) Describe the importance of this type of bond named in **(a)(i)** in **R**.

....

....

....

.... [2]

(b) State **one** role of **P** in translation.

....

.... [1]

For examiner's use

Not all mRNAs which enter the cytoplasm are used immediately for translation. Initiation of translation is prevented because some mRNAs lack poly-A tails of sufficient length. At the appropriate time, a polyadenylation enzyme will add more adenine nucleotides to lengthen the poly-A tail, allowing translation of these mRNAs to begin.

(c) Describe the mode of action of the polyadenylation enzyme.

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.... [4]

In the cell, free adenine nucleotides that are added to the poly-A tail exist as adenosine triphosphate. A molecule of adenosine triphosphate is shown in Fig. 2.2a below.

Introduction of Cordycepin into the cell prevents the polyadenylation enzyme from carrying out its function at the 3' end of mRNA. Translation does not begin in the presence of Cordycepin. Fig. 2.2b shows the structure of Cordycepin.

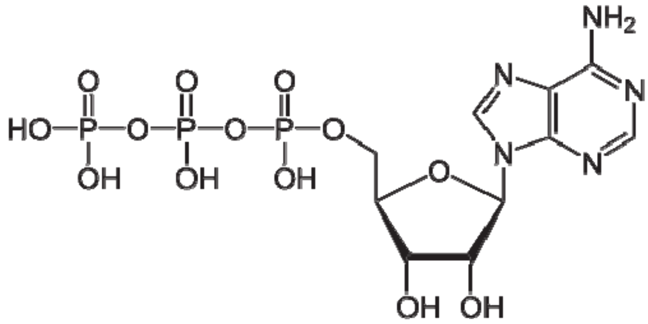

Fig. 2.2a

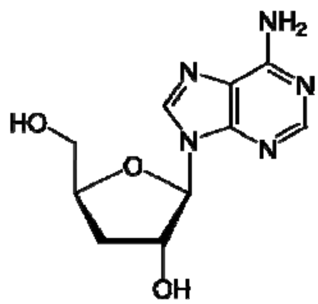

Fig. 2.2b

(d) Suggest how Cordycepin prevents translation of mRNA in the cytoplasm.

....

....

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.... [2]

[Total: 10]

For examiner's use

3 In a species of bird, feathers may be white, black or black with white spots (spotted). In this same species, beaks may be long or short.

In a study to check on genetic diversity of this species of bird, a research ecologist examined the results obtained from the crosses of two pairs of birds. The results of the crosses are shown in Table 3.1 below.

Table 3.1

	Cross	**Ratio of F1 phenotypes**
1	Black bird with long beak crossed with spotted bird with long beak	3 black with long beak : 3 spotted with long beak : 1 black with short beak : 1 spotted with short beak
2	Spotted bird with long beak crossed with white bird with short beak	1 spotted with long beak: 1 white with long beak

(a) Using a genetic diagram, explain the results of the cross between a black bird with long beak and a spotted bird with long beak as shown in Table 3.1.
.

[4]

For examiner's use

(b) Describe the cross you would carry out to determine if a randomly picked F_1 offspring with black feathers and long beak from the cross in part (a) is homozygous for beak length.

[2]

In order to confirm the genotype of the above F_1 offspring, polymerase chain reaction (PCR) was carried out on a DNA sample of the bird. The investigator added a mixture to the DNA sample and in a few hours, the tiny sample of DNA was amplified millions of times.

(c) Identify the components in the mixture that was added with the DNA sample and their respective functions for PCR to amplify the sample of DNA present.

[3]

(d) Explain **one** possible limitation of this procedure.

[1]

[Total: 10]

For examiner's use

4 Primate species have the potential to visually perceive three colour spectrums – blue, green and red.

Fig. 4.1 shows a phylogenetic tree of primate evolution. The common ancestor for all primate species was thought to be dichromatic, being able to perceive blue and one other colour only (hence unable to distinguish green and red colour). Only the species within the group of "apes" and "old world monkeys" are found to be consistently trichromatic and are able to perceive all three colours.

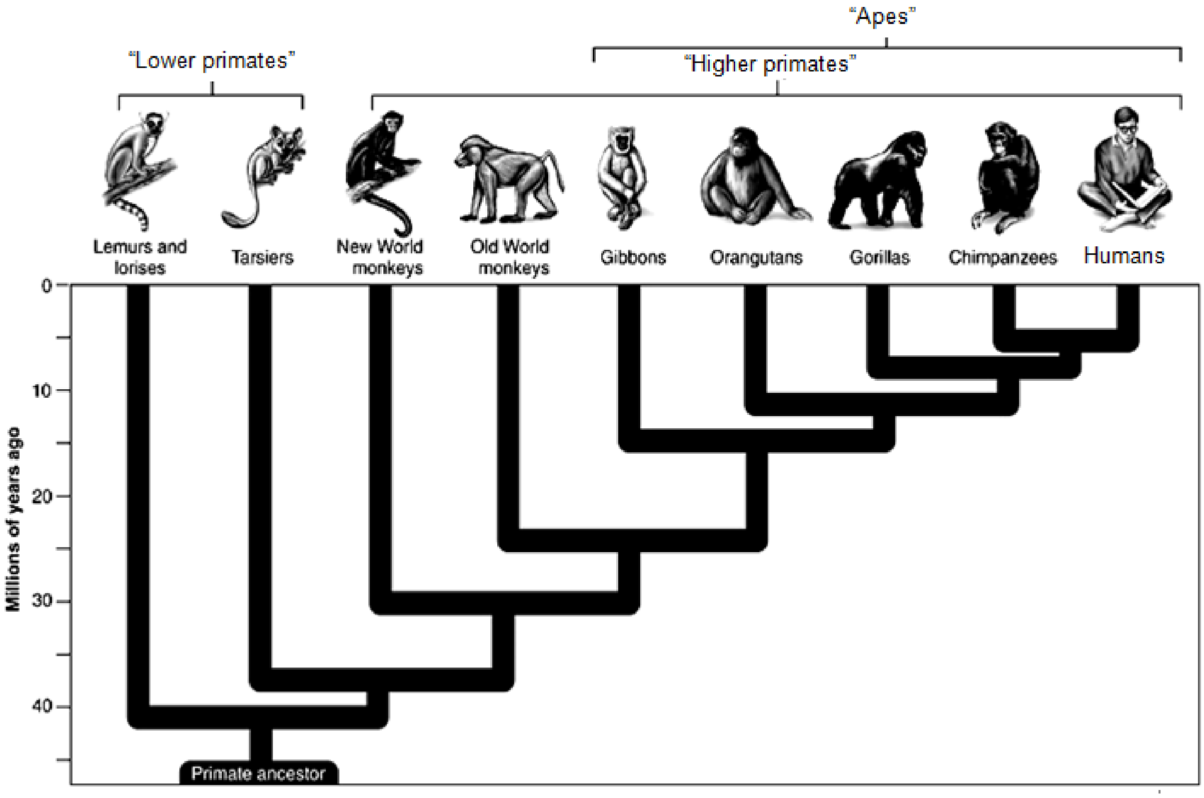

Fig. 4.1

(a) (i) Using an arrow, indicate on Fig. 4.1 the point where trichromatic vision is likely to have evolved in primates. [1]

(ii) Suggest how trichromatic vision may be of survival value to fruit-feeding primates in forest habitats.

....

.... [1]

(ii) Explain how natural selection has resulted in the evolution of trichromatic vision in a population.

[3]

Almost all known species of "new world monkeys" have dichromatic vision. The gene for blue colour vision (gene B) is found on chromosome 7, while green colour vision and red colour vision are controlled by different alleles of the same gene (gene G/R) found on chromosome X.

Howler monkeys (*Alouatta* spp.) are a genus of "new world monkeys" with trichromatic vision. They share the same gene B with other species of "new world monkeys". However, on chromosome X, the gene for green colour vision (gene G) is found on a different locus than the gene for red colour vision (gene R). Both gene G and gene R may therefore coexist on a single chromosome.

(b) Describe the mutation that could have led to trichromatic vision in howler monkeys.

[2]

Cone cells are somatic cells found in the retina of the eye which are responsible for colour vision.

(c) (i) Describe **two** differences between cone cells and adult stem cells.

[2]

(ii) State the function of adult stem cells in the body.

[1]

[Total: 10]

For examiner's use

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

5 **(a)** Describe the structure and functions of lysosome. [6]

(b) Describe how variations may arise in a population and explain their role in evolution. [8]

(c) With reference to Bt. corn, explain the significance of genetic engineering to solve the global demand for food. [6]

[20]

OR

6 **(a)** Compare and contrast the structures of glycogen and a globular protein. [8]

(b) Explain the production of a small yield of ATP from anaerobic respiration in mammalian cells. [6]

(c) Outline the procedure for cloning a eukaryotic gene with the use of bacterial plasmid. [6]

[20]